

A digital solution for informed dietary choices

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Abstract:

We have identified a social problem related to the preference for buying ultra-processed foods rather than cooking at home. Ideally, we should opt for consuming minimally processed foods such as fresh vegetables, wholegrain rice, oats, and nuts.

Through research and data available on government websites and scientific papers, we have compiled a list of essential guidelines for consumers to consider when choosing what to eat. Our focus is on highlighting the scientifically proven long-term health effects of additives and guiding consumers to select food items with as few additives as possible. Furthermore, we have taken into account the macronutrient composition, advocating for the avoidance of food items abundant in trans fats, sodium, and sugars.

Key Words: Nutrition, Health, Additives, Macronutrients, Choice, Awareness

1. Health issues

1.1. Causes of death. Main statistical results

According to Harvard Health, people often worry about the wrong things. Statistics show people worry more about things like plane crashes and shark attacks, while the top killers are heart disease and cancer.¹

The latest estimated information on causes of death is available for the reference period 2012. Fig. 1. shows that circulatory diseases and cancer (malignant neoplasms) were by far the leading causes of death in the EU.

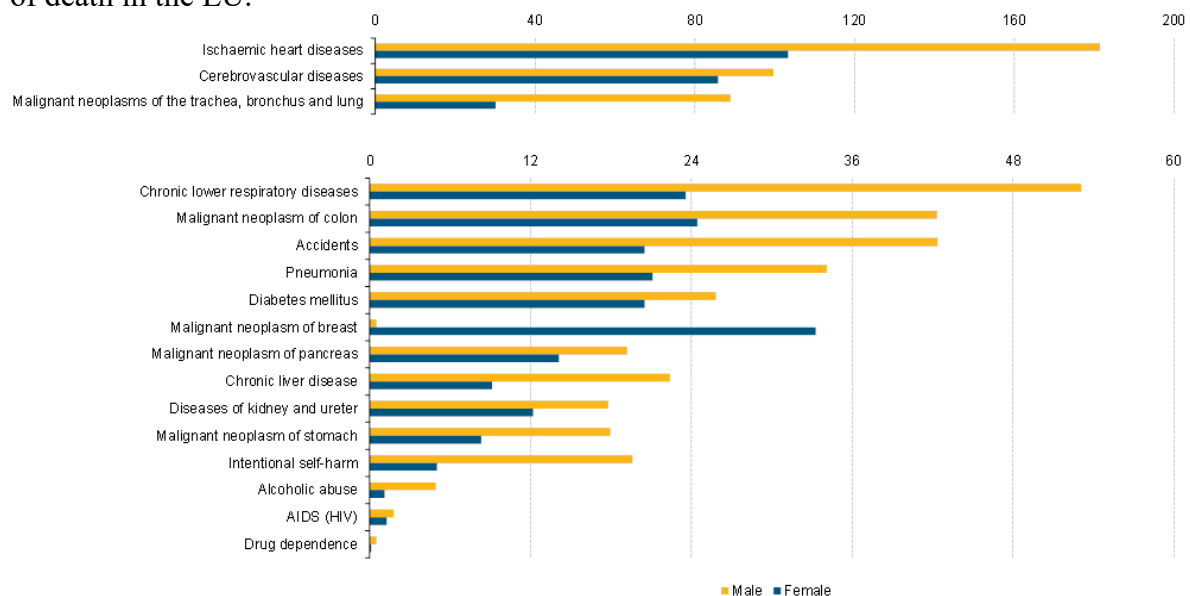


Fig. 1. Causes of death, standardised death rate per 100.000 inhabitants
[eurostat, Statistics Explained, [https://ec.europa.eu/eurostat/statistics-explained/images/Causes_of_death_-_standardised_death_rate_\(per_100_000_inhabitants\).png](https://ec.europa.eu/eurostat/statistics-explained/images/Causes_of_death_-_standardised_death_rate_(per_100_000_inhabitants).png)]

¹ Shmerling, R. H., "How well do you worry about your health?", Harvard Health Publishing, www.health.harvard.edu, 2024

According to the National Institute of Statistics Romania, the main causes of death in 2021-2022 have been:

- 1) Diseases of the circulatory system(686.5/100 000 inhabitants)- include hypertensive diseases, atherosclerosis(stiffening of the arteries- one of the reasons is the accumulation of a lipid formation, mainly cholesterol):
 - a) arterial hypertension
 - b) cerebrovascular diseases (most common: stroke)
 - c) other ischemic heart diseases
 - d) acute myocardial infarction
 - e) chronic pulmonary heart: change in the structure and function of the heart due to a primary disease of the respiratory system
- 2) Respiratory diseases
- 3) Tumors:
 - a) respiratory system
 - b) breast
 - c) stomach
 - d) prostate
 - e) cervix
- 4) Diseases of the digestive system
- 5) Injuries, poisoning, and other consequences of external causes (e.g. traumatic injuries due to road accidents, burns, drug, and medication intoxication)

1.2. Prevention

According to Medlife², methods of prevention of cardiovascular diseases are:

- 1) Mediterranean diet and avoidance of excess calories
- 2) 150 min moderate physical activity/week: minimum 7000 steps/day³
- 3) Quit smoking
- 4) Reduce high cholesterol (possibly also through a diet reduced in unsaturated fats) and hypertension
- 5) Ensuring mental health - reducing stress
- 6) Usual blood tests

According to the World Health Organization⁴, only 42% of adults with high blood pressure are diagnosed and receive treatment, leaving a significant portion of the population at risk.

Alarmingly, just 1 in 5 adults with hypertension have their condition under control.

Uncontrolled high blood pressure is a major contributor to premature deaths around the world.

Despite global efforts to reduce hypertension by 33% since 2010, a significant gap remains in diagnosing and managing this critical health condition.

Recommendations- don't:

- eat too much salty food (try to stay under 2 grams per day)
- eat foods high in saturated or trans fats
- smoke or use tobacco
- drink too much alcohol (1 drink daily max for women, 2 for men)

² Vlădoianu, A., “Bolile cardiovasculare: tipuri, cauze, diagnostic, tratament și prevenție”, www.medlife.ro, 2022

³ Williamson, L., “Walking your way to better health? Remember the acronym FIT”, www.heart.org, 2022

⁴ World Health Organisation, WHO, www.who.int

1.3. Risk factors for primary hypertension

- 1) Family history: If a close relative has hypertension, you are more likely to develop it.
- 2) Age: The risk of hypertension increases with advancing age.
- 3) Obesity or overweight: Excess weight increases pressure on the heart and blood vessels.
- 4) Sedentary lifestyle: Lack of regular physical activity increases the risk of hypertension.
- 5) Unhealthy diet: Excessive consumption of salt, saturated fat, and sugar can contribute to high blood pressure.
- 6) Alcohol consumption: Alcohol can temporarily increase blood pressure.
- 7) Smoking: Smoking damages blood vessels and increases blood pressure.
- 8) Stress: Chronic stress can contribute to high blood pressure.

(Dimsdale, 2008)⁵ clarifies how we define stress and explores the difference between short-term and long-term stress on the heart.

Examples are given of how sudden events (earthquakes) and ongoing pressures (work stress, unhappy marriage) can impact the heart.

Research shows stress contributes to various heart problems, including sudden death, heart attacks, and irregular heartbeats.

While stress seems to trigger these events, it's not entirely clear if it directly causes them.

There's strong evidence that stress can harm the heart, but individual factors can make people more or less susceptible.

In conclusion, the article says that short-term stress is inevitable and sometimes needed, but if it transforms into long-term stress, then it can have adverse effects on the heart.

1.4. The connection between nutrition and disease

Even with advances in treating it, cardiovascular disease is still a major health issue.

Diet is the most crucial lifestyle factor affecting heart health globally.

Traditionally, focusing on individual nutrients (cholesterol hypothesis) wasn't as effective as focusing on overall dietary patterns.

Recent studies on the Mediterranean diet showed success in reducing CVD risk. This paper examines the evidence for using dietary interventions, specifically focusing on how diet can prevent and improve coronary artery disease and heart failure.

The conclusions of the study (Ravera, et al., 2016)⁶:

Focusing on overall diet in managing heart disease (CVD) has proven to be a successful strategy with clear benefits. These studies suggest that combining foods into specific dietary patterns creates a stronger effect than focusing on individual nutrients. In the future, additional research trials might shed more light on how different food combinations impact hard cardiovascular outcomes (serious heart problems).

1.5. The connection between salt and high blood pressure

Lots of research shows a link between how much salt you eat and your blood pressure.

Most people eat too much salt (around 9-12 grams a day).

⁵ Dimsdale, J. E., „Psychological Stress and Cardiovascular Disease”, Journal of the American College of Cardiology, www.jacc.org, 2008

⁶ Ravera A.; Carubelli V.; Sciatii E.; Bonadei I.; Gorga E.; Cani D.; Vizzarda E.; Metra M.; Lombardi C., „Nutrition and Cardiovascular Disease: Finding the Perfect Recipe for Cardiovascular Health”, www.mdpi.com, 2016

Doctors recommend aiming for 5-6 grams of salt daily, which helps lower blood pressure for everyone.

Eating even less salt (3-4 grams) offers even greater benefits.

Studies suggest less salt may also reduce your risk of heart disease and heart failure.

For people with heart failure already, high salt intake can worsen symptoms. Lower salt intake is crucial for managing this condition.

The ideal amount of salt reduction for heart failure patients is unclear, but they likely benefit from following the general recommendation of less than 5 grams daily.⁷

2. Ultra-processed foods

2.1. How to identify them

(Monteiro, et al., 2019)⁸ describes a food classification system called NOVA and explains how to identify ultra-processed foods, which are linked to various health problems.

The NOVA system categorizes foods based on the extent and purpose of industrial processing. It has 4 groups:

- Group 1: Unprocessed or minimally processed foods (fruits, vegetables, whole grains)
- Group 2: Processed culinary ingredients (oils, sugar, salt)
- Group 3: Processed foods (cheese, canned vegetables)
- Group 4: Ultra-processed foods (explained below)

Ultra-processed foods: These are highly processed products made with ingredients rarely used in home cooking and lots of additives. They are designed to be profitable, convenient, and hyper-palatable. Examples include sugary drinks, packaged snacks, instant noodles, and frozen meals.

Health concerns: Studies suggest diets high in ultra-processed foods are linked to obesity, heart disease, diabetes, and other chronic illnesses.

Identifying ultra-processed foods: Look at the ingredients list. If it includes items unfamiliar for home cooking (like high-fructose corn syrup, and hydrolyzed proteins) or many additives (like flavors, colors, emulsifiers), it's likely ultra-processed.

The article concludes by emphasizing that ultra-processed foods are not "real food" and that avoiding them is crucial for a healthy diet.

2.2. About NOVA

In the report "The UN Decade of Nutrition, the NOVA food classification and the trouble with ultra-processing"⁹, the authors advocate for the adoption of a system of grades from 1 to 4 to allow to simply compare the degree of processing of products.

New research associating researchers from Inserm, Inra, and the Paris 13 University (Centre de recherche épidémiologie et Statistique Sorbonne Paris Cité, équipe EREN) suggests a correlation between the consumption of ultra-transformed foods and an increased risk of developing a cancer.

⁷ He F.J., Burnier M., MacGregor G.A., „Nutrition in cardiovascular disease: salt in hypertension and heart failure”, academic.oup.com, 2011

⁸ Monteiro, C.A.; Cannon, G.; Moubarac, J.C; Bertazzi-Levy, R.; Louzada, M.L.C.; Constante-Jaime, P., „Ultra-processed foods: what they are and how to identify them”, www.cambridge.org, 2019

⁹ Monteiro, C.A.; Cannon, G.; Moubarac, J.C; Bertazzi-Levy, R.; Louzada, M.L.C.; Constante-Jaime, P., „The UN Decade of Nutrition, the NOVA food classification and the trouble with ultra-processing”, <https://pubmed.ncbi.nlm.nih.gov>, 2017

2.3. Group 3 (NOVA): Processed Foods

Made by adding ingredients from Group 2 (oils, sugar, salt) to Group 1 foods (vegetables, fruits, grains).

Examples: canned fish, bottled vegetables, fruits in syrup, cheeses.

Processing methods include preservation (canning, bottling), cooking, and fermentation (for bread and cheese).

Usually have 2-3 ingredients and are easily recognizable as modifications of whole foods (fruits, vegetables, grains).

It can be eaten alone or combined with other foods.

The main purposes of processing here is to:

- Increase the shelf life of unprocessed foods.
- Modify or improve the taste, texture, or other sensory qualities of the original food.

2.4. Group 4 (NOVA): Ultra-processed foods

They are formulations made from refined ingredients and additives, not whole foods. Here's a breakdown of what makes them different:

- Ingredients:

Rarely used at home for cooking (e.g. high-fructose corn syrup, hydrolyzed proteins)

Include additional sources of energy and nutrients not typical in cooking (e.g. casein, whey protein)

It may contain further processed ingredients like hydrogenated oils or maltodextrin

- Additives:

Include standard preservatives and stabilizers found in processed foods

Uniquely contain additives to improve taste or hide unpleasant aspects (e.g. artificial flavors, colors, sweeteners)

- Processing:

It involves many steps often not done at home (e.g. hydrogenation, extrusion)

- Overall purpose:

To be convenient, shelf-stable, appealing, and profitable

- Examples: sugary drinks, packaged snacks, frozen meals
- They are heavily marketed and packaged to be attractive

3. What is happening worldwide according to The World Health Organization

Here are the key points of the article "Children and young people must be at the heart of food systems transformation"¹⁰:

- Problem: Many children suffer from malnutrition due to poor diets and inaccessible nutritious foods.
- Causes: Broken food systems, poverty, and unhealthy processed food marketing contribute to malnutrition in children.
- Solution: Transform food systems to prioritize the needs of children.
- Key actions:
 - Make healthy foods affordable and accessible through subsidies and pricing.
 - Improve food quality through fortification and reduction of unhealthy ingredients.
 - Promote healthy eating habits through education and marketing regulations.

¹⁰ World Health Organisation, „Children and young people must be at the heart of food systems transformation”, www.who.int, 2021

Implement clear food labeling.

Create healthy food environments in schools and communities.

- Goal: Ensure every child has access to safe, nutritious, and affordable food for a healthier future.
- Collaboration: Stakeholders like governments, families, and young people need to work together to achieve this goal.

Food safety, nutrition, and food security are closely linked. Unsafe food creates a vicious cycle of disease and malnutrition, particularly affecting infants, young children, the elderly, and the sick. In addition to contributing to food and nutrition security, a safe food supply also supports national economies, trade, and tourism, stimulating sustainable development. The globalization of food trade, a growing world population, climate change and rapidly changing food systems have an impact on the safety of food.

4. Food Additives

The Royal Society of Chemistry recommends "Essential Guide to Food Additives" as a reliable source of information in a field often clouded by emotions rather than scientific evidence.

Classes of additives:

- | | |
|--------------------------------|---------------------|
| 1. Preservatives | 5. Sweeteners |
| 2. Antioxidants | 6. Flavor enhancers |
| 3. Emulsifiers and stabilisers | 7. Flavorings |
| 4. Colors | 8. Other additives |

Preserving food is a long-standing challenge:

People have been using techniques like drying, salting, and pickling for centuries to keep food safe throughout the year.

Technologies like canning, refrigeration, and freezing are relatively recent inventions.

The focus shifted towards making food more appealing in taste and appearance.

Two main purposes:

- 1) Preserving food safety: additives prevent spoilage by bacteria, oxidation, and other chemical changes;
- 2) Additives improve the taste, appearance, and texture of food;

Concerns: globalization and capitalism have fueled the use of food additives by increasing demand for processed foods, fostering market competition and profit maximization, promoting standardization and homogenization of food products, and facilitating access to ingredients and technology.

4.1. Roles of each class of additive

Preservatives

- Used to counteract the perishability of food and extend the shelf-life
- The main benefit is that they prevent the growth of harmful bacteria that can cause food poisoning

Antioxidants

- They slow down spoilage caused by oxygen, which can make fats turn bad (rancid), lose flavor and color, and even lose nutrients

Emulsifiers and stabilisers

- Emulsifiers are added to processed foods like mayonnaise, ice cream, chocolates, peanut butter, cookies, creamy sauces, margarine, and baked goods to prevent the separation of their oil and water components

Colours

- Enhances the visual properties of foods
- Controversial because is perceived as a means of deceiving the consumer about the nature of food
- Most brightly colored products are those aimed at children

Sweeteners

- Sugar substitutes offer health benefits such as reduced risk of tooth decay, stable blood sugar levels, and potential short-term aid in weight management for those with overweight or obesity.
- Sugar substitutes, deemed safe by health agencies, are not linked to cancer. While some studies suggest a potential association with higher risks of stroke and heart disease, it's unclear if this is directly caused by sugar substitutes. They may affect gut health and cause gastrointestinal symptoms in some individuals. Moderation in consumption is advised, especially for long-term use.¹¹

Flavor enhancers

- Monosodium Glutamate (MSG) is the most popular flavour enhancer and is widely used in commercial foods, providing a distinct umami flavor.
- MSG has been linked to various health issues including obesity, metabolic disorders, and neurotoxic effects. Despite its flavor-enhancing properties, its association with toxicity prompts consideration for public health safety.
- The debate revolves around whether measures to prevent harm should be implemented or if a complete ban on the product is warranted.¹²

Flavourings

- They impart taste and odor to foods. Flavouring substances are categorized as natural, nature identical, or artificial, with around 2700 substances identified and registered in the European Union.
- Are often complex mixtures protected by commercial confidentiality.
- Labels typically list "flavourings" due to the complexity and length of individual substances.
- Stricter regulations on natural flavouring labeling are proposed.

Other additives

- Colors and sweeteners have specific legislation due to their function, while other additives are classified as "miscellaneous."
- Miscellaneous additives include thickeners, acids, acidity regulators, anti-caking agents, anti-foaming agents, bulking agents, carriers, glazing agents, humectants, raising agents, and sequestrants.
- Thickeners, often starch- or gum-based, are commonly used to adjust the texture and viscosity of food and drinks.
- Packaging gases, like those used in pre-packed products, modify the atmosphere within the pack to prolong shelf life.¹³

¹¹ Mayo Clinic Staff, „Artificial sweeteners and other sugar substitutes”, www.mayoclinic.org, 2023

¹² Niaz K., Zaplatic E., Spoor J., “Extensive use of monosodium glutamate: A threat to public health?”, www.ncbi.nlm.nih.gov, 2018

¹³ books.google.ro

5. The identified problem we are trying to address

According to the extensive medical sources cited previously, nutrition profoundly impacts our health, representing a lifestyle choice rather than a predetermined circumstance to endure or accept. Each day presents an opportunity for us to make more informed decisions regarding nutrition if we avail ourselves of accurate information.

Nevertheless, the prevailing challenge lies within our fast-paced, constantly on-the-go society, where a meticulous selection of food products may inadvertently take a back seat to more convenient options such as pastries, fast-foods, and attractively packaged, ready-to-eat sandwiches boasting lengthy expiration dates—a worrisome indication. Amidst numerous obligations, the meticulous consideration of exact ingredients and their long-term effects on health may often be neglected.

Acknowledging the Sisyphean task of scrutinizing every label for every item within one's basket, particularly when confronted with incomprehensible chemical terminology, we recognized the need for a digital solution to streamline this selection process. This digital solution aims to identify lesser-known ingredients based on research findings while elucidating the health implications of each dubious component. Furthermore, it categorizes these questionable ingredients (additives) into three distinct tiers—red, orange, and yellow—based on the currently known data related to potential long-term health issues.

6. Our solution

6.1. Concept

With meticulous research into additives and unhealthy chemical substances, HealthyTurtle has been developed to bring awareness to the forefront when ordering food online. Our extension offers educational insights into identified additives and chemicals, empowering consumers to make informed decisions about their food choices.

HealthyTurtle aims to be an easy-to-use tool that helps users become more aware of the contents of the food they purchase from online stores. By leveraging the potential of HealthyTurtle, consumers can confidently navigate online food platforms, knowing they have a vigilant companion safeguarding their health and well-being.

To provide ease of use, HealthyTurtle is available as a browser extension, ensuring quick access when browsing the web. Developed for Google Chrome, the extension seamlessly integrates with the popular browser. The online store chosen for this project is Freshful¹⁴, a store made popular by its parent company, eMAG.

We have examined each food item from two distinct perspectives: one concerning the added chemical substances commonly present in highly processed food, and the other about macronutrients, including carbohydrates, proteins, fats, and fiber.

In our assessment, we have considered the established classification of fats into "good" and "bad" categories: specifically, unsaturated fats are regarded as beneficial, while trans and saturated fats are deemed detrimental. Notably, trans fats pose an increased risk of disease, even when consumed in small quantities.¹⁵

¹⁴ <https://www.freshful.ro/>

¹⁵ www.hsph.harvard.edu/nutritionsource

6.2. Presentation and design

HealthyTurtle's icon is a cartoon turtle visible in Fig. 2. Both the name and the icon were chosen for their friendly nature. The icon was found on Pinclipart¹⁶.

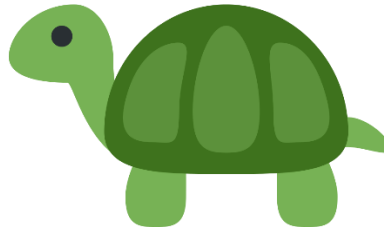


Fig. 2. HealthyTurtle icon
www.pinclipart.com

The interaction menu is a simple pop-up page featuring two headings and a list of common food ingredients, each with a dropdown next to them that allows the user to select between three quantity options. The ingredients are salt, sugar, gluten, lactose, eggs, fish, soy and nuts. The options are "Indifferent", "Reduced" and "None".

HealthyTurtle

Options

Salt:	None
Sugar:	None
Gluten:	Reduced
Lactose:	Reduced
Eggs:	None
Fish:	Reduced
Soy:	Reduced
Nuts:	None
Save	

Fig. 3. Pop-up menu

The highlighting of the ingredients is done by changing the background color of the words on the page: orange, if the option is "Reduced" and red if the option is "None".

Lapte de vacă semidegresat și pasteurizat proteine din lapte culturi
este o enzimă care descompune lactoza în glucoză și galactoză

Fig. 4. Highlighting example

The pop-up page is an HTML file containing two headings, a table, and a button. The table is made up of eight rows, each of them containing an ingredient's name and a <select> element (the dropdown) containing three <option> elements, one for each of the options previously presented. The button is a "Save" button that triggers a JavaScript function for storing the options in memory.

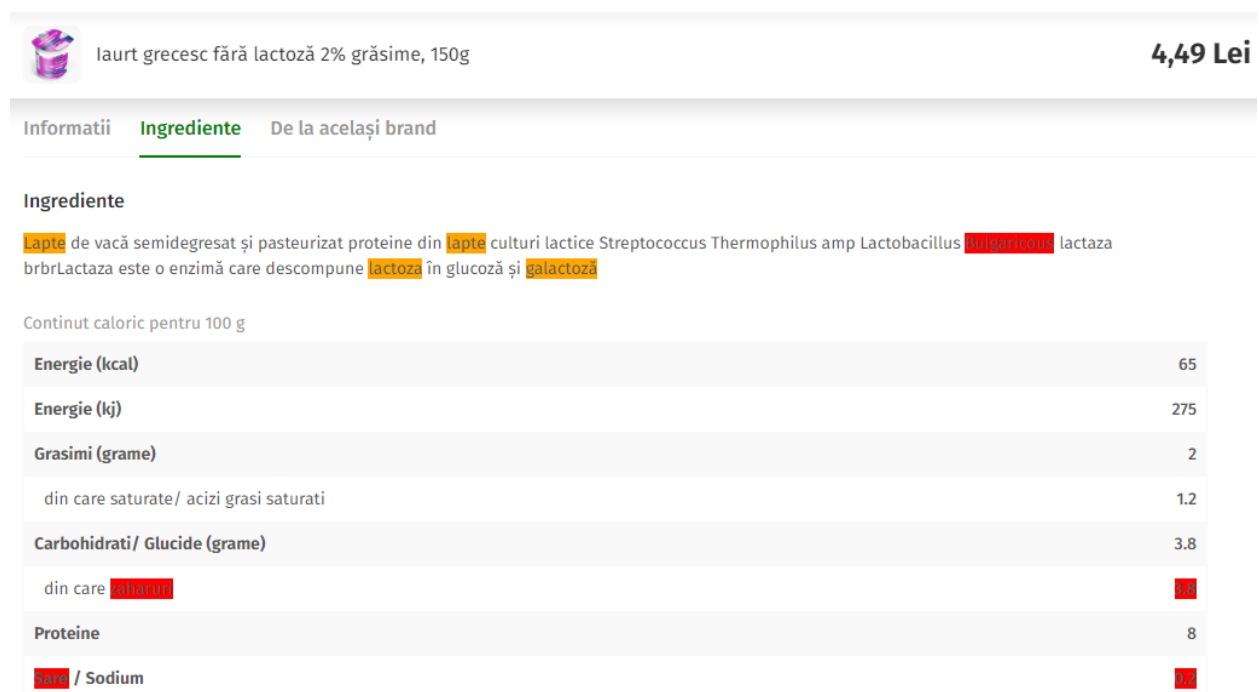
¹⁶ www.pinclipart.com

6.3. Functionality

Once the user clicks on the extension, the pop-up page appears which, upon loading, triggers a JavaScript function from the "options.js" script called "restoreOptions" that uses the chrome.storage API to get the user's options and update each of the <select> element's value to the one stored in memory.

Whenever the "Save" button is clicked, another JS function is triggered, "saveOptions", which also uses Chrome.storage API, but this time to store each <select> element's value. Once the saving is done, a message reading "Options saved." is briefly displayed.

A script called "script.js" is injected into the page. As soon as the user clicks on the "Ingrediente" tab of a product, the script loads into memory the user's options and begins verifying each word in the text area of the tab, as well as in the "Cotinet caloric pentru 100g" table, to see if they represent an ingredient from the list and, if so, it wraps the word inside a with the "style" attribute that sets its background-color property to a color appropriate to the user's choice, should the choice not be "Indifferent".



The screenshot shows a product page for 'Iaurt grecesc fără lactoză 2% grăsime, 150g' priced at 4,49 Lei. The 'Ingrediente' tab is selected, showing a list of ingredients: 'Lapte de vacă semidegresat și pasteurizat', 'proteine din lapte', 'culturi lactice Streptococcus Thermophilus amp Lactobacillus bulgaricus', and 'lactaza'. A note explains that lactase is an enzyme that breaks down lactose into glucose and galactose. Below the ingredients is a table titled 'Cotinet caloric pentru 100 g' with the following data:

Componentă	Valoare
Energie (kcal)	65
Energie (kj)	275
Grasimi (grame)	2
din care saturate/ acizi grasi saturati	1.2
Carbohidrati/ Glucide (grame)	3.8
din care zaharuri	3.8
Proteine	8
sare / Sodium	0.2

Fig. 5. Result

6.4. Future improvements

At the time of writing, the goal of having the extension highlight other ingredients that the user might not be aware of has not been achieved. Many of the databases considered for this purpose contained plenty of information, however, the data was too raw for us to be able to process it inside of a program. As such, a .csv file was created. It contains information carefully extracted from the aforementioned databases about the most common dangerous compounds found in food. This data would have been used to give these compounds the same treatment, however, the implementation could not be finished in time, but is a top priority for the future.

Another future improvement is deeper personalisation by processing and using more information about the user such as age, sex, predispositions and other affections. All of these would be used to

warn the user that they should, perhaps, think twice before buying a certain item. For example, the extension could warn the user that certain foods involve a higher risk past a certain age or that certain ingredients aren't good for the liver, should they have an affection of that manner. This way, even uninformed or inexperienced people could not only benefit more from HealthyTurtle, but also learn valuable facts that could be relevant in their daily lives. This could be achieved via custom tooltips for each of the highlighted elements, making it so that when the user hovers their mouse over them, information could be displayed.

A certain improvement would be to allow users to input "custom" options: ingredients that aren't already in the extension. This feature would ideally use a search API that could check the user's spelling, similar to Google Search. For the more meticulous of users, an "advanced" mode would be made, one that, when toggled on, would allow the user to input exact minimum and maximum values for each ingredient per 100 grams.

HealthyTurtle could also use some styling, which it currently lacks. A common option to implement would be "Dark Mode" which turns the colours of the interface from bright to dark. Perhaps, a selection of themes could be made, one that would contain more "nuanced" options such as "Neon" or "Comet Blue". The ability to set a custom background for the pop-up would also be worth considering.

A final and most useful improvement would be to "broaden the scope" of the extension such that it could be used on all the items on a page, instead of individual items. It would colour code each product based on the user's preferences and the chemical contents. This way, the user's search for healthier products would be much faster, easier and more convenient.

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